of the wave. The Polaroid P_1 acts as a polarizer and the transmitted beam is linearly polarized. The second Polaroid acts as an analyzer.

22.2 MALUS' LAW LO 1

Let us consider a Polaroid P_1 which has a pass-axis parallel to the x-axis (see Fig. 22.6); i.e., if an unpolarized beam propagating in the z direction is incident on the Polaroid, then the electric vector associated with the emergent wave will oscillate along the x-axis. We next consider the incidence of the x-polarized beam on the Polaroid P_2 whose pass axis makes an angle θ with the x-axis (see Fig. 22.6). If the amplitude of the incident electric field is E_0, then the amplitude of the wave emerging from the polaroid P_2 will be $E_0 \cos\theta$ and thus the intensity of the emerging beam will be given by

$$I = I_0 \cos^2\theta \qquad \textbf{(22.9)}$$

where I_0 represents the intensity of the emergent beam when the pass axis of P_2 is also along the x-axis (i.e., when $\theta = 0$). Equation (22.9) represents Malus' Law. Thus, if a linearly polarized beam is incident on a Polaroid and if the Polaroid is rotated about the z-axis, then the intensity of the emergent wave will vary according to the above law. For example, if the Polaroid P_2 shown in Fig. 22.6 is rotated in the clockwise direction, then the intensity will increase till the pass-axis is parallel to the x-axis; a further rotation will result in a decrease in intensity till the pass-axis is parallel to the y-axis, where the intensity will be almost zero. If we further rotate it, it will pass through a maximum and again a minimum before it reaches its original position.

Figure 22.7 shows actual photographs of two Polaroids at different relative orientations. In Fig. 22.7 (a) the two are parallel to each other and therefore almost the entire light passes through. In Fig. 22.7 (b) the two Polaroids are oriented at 45° with respect to each other and about 50% of the light passes through; because according to Malus' law $I = I_0 \cos^2 45 = \frac{1}{2} I_0$. Finally, in In Fig. 22.7 (c) the two Polaroids are at right angles to each other (notice the position of the blue dot) and almost no light passes through because $I = I_0 \cos^2 90^\circ = 0$ [see also Fig. 26 in the prelim pages of the book]

22.3 PRODUCTION OF POLARIZED LIGHT LO 1

In this section we will discuss various methods for producing linearly polarized light waves.

22.3.1 The Wire Grid Polarizer and the Polaroid

The physics behind the working of the wire grid polarizer is probably the easiest to understand. It essentially consists of a large number of thin copper wires placed parallel to each other as shown in Fig. 22.8. When an unpolarized electromagnetic wave is incident on it, then the component of the electric vector along the length of the wire is absorbed. This is due to the fact that the electric field does work on the electrons inside the thin wires and the energy associated with the electric field is lost in the Joule heating of the wires. On the other hand, (since the wires are assumed to be very thin) the component of the electric vector along the x-axis passes through without much attenuation. Thus, the emergent wave is linearly polarized with the electric vector along the x-axis. However, for the system to be effective (i.e., for the E_y component to be almost completely attenuated) the spacing between the wires should be less than λ. Clearly, the fabrication of such a polarizer for a 3 cm microwave is relatively easy because the spacing has to be less than 3 cm. On the other hand, since the light waves are associated with a very small wavelength $(\sim 0.5\ \mu\text{m})$, the fabrication of a polarizer in which the wires are placed at distances less than $0.5\ \mu\text{m}$ is extremely difficult. Nevertheless, Bird and Parrish did succeed in putting about 30,000 wires in about one inch; for further details see Refs. 22.1 and 22.2. The details of the procedure for making this wire grating are also discussed in Ref. 22.1. The original work of Bird and Parrish was published in 1950 (see Ref. 22.2).

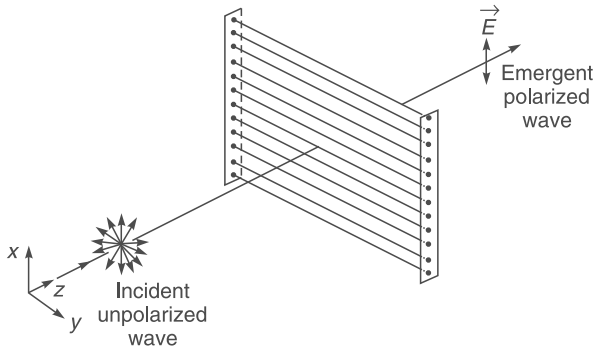


Fig. 22.8 The wire-grid polarizer.

As already pointed out, it is extremely difficult to fabricate a wire grid polarizer which would be effective for visible light. However, instead of long thin wires, one may employ long chain polymer molecules that contain atoms (like iodine) which provide high conductivity along the length of the chain. These long chain molecules are aligned so that they are almost parallel to each other. Because of the high conductivity provided by the iodine atoms, the electric field

parallel to the molecules get absorbed. A sheet containing such long chain polymer molecules (which are aligned parallel to each other) is known as a **Polaroid**. When a light beam is incident on such a Polaroid, the molecules (aligned parallel to each other) absorb the component of electric field which is parallel to the direction of alignment because of the high conductivity provided by the iodine atoms; the component perpendicular to it passes through. Thus, the aligned conducting molecules act similar to the wires in the wire grid polarizer and since the spacing between two adjacent long chain molecules is small compared to the optical wavelength, the Polaroid is usually very effective in producing linearly polarized light. The aligning of the long chain conducting molecules is not very difficult and the experimental details of producing the polarizer are given in Ref. 22.1.

22.3.2 Polarization by Reflection

We consider the incidence of a plane wave on a dielectric; we assume that the electric vector associated with the incident wave lies in the plane of incidence as shown in Fig. 22.9 (a). It will be shown in Sec. 24.2 that if the angle of incidence θ is such that

$$\theta = \theta_p = \tan^{-1}\left(\frac{n_2}{n_1}\right) \qquad \textbf{(22.10)}$$

then the reflection coefficient is zero. Thus, if an unpolarized beam is incident at this angle, then the reflected beam will be linearly polarized with its electric vector perpendicular to the plane of incidence [see Fig. 22.9(b)]. Equation (22.10) is referred to as the *Brewster's law* and at this angle of incidence, the reflected and the transmitted rays are at right angles to each other; the angle θ_p is known as the polarizing angle or the Brewster angle.

A (commercially available) polarized sunglass blocks the horizontal component and allows only the vertical component to pass through [see Fig. 22.10(a)]. For the air-water interface, $n_1 \approx 1$ and $n_2 \approx 1.33$ and the polarizing angle $\theta_p \approx 53^\circ$. Thus, if the sunlight is incident on the sea at an angle close to the polarizing angle, then the reflected light will be almost linearly polarized [see Fig. 22.10 (b)] and if we now wear polarized sunglasses, the glare, i.e., the light reflected from the water surface, will not be seen. This is the reason why polarized sunglasses are often used by fishermen to remove the glare on the surface and see the fish inside water. Figure 22.11 (a) shows a photograph on the road with ordinary glasses; if we use polarized lenses, the glare can be considerably reduced as shown in Fig. 22.11 (b); see 27 and 28 in the prelim pages. Figure 22.12 shows sunlight incident on a water surface at an angle close to the polarizing angle so that the reflected light is almost polarized. If the Polaroid allows the (almost polarized) reflected beam to pass through, we see the glare from water surface [see Fig. 22.12 (a)]; the glare can be blocked by using a vertical polarizer and one can see the inside of the water [see Fig. 22.12 (b) and Figs. 29 (a) and (b) in the prelim pages of the book].

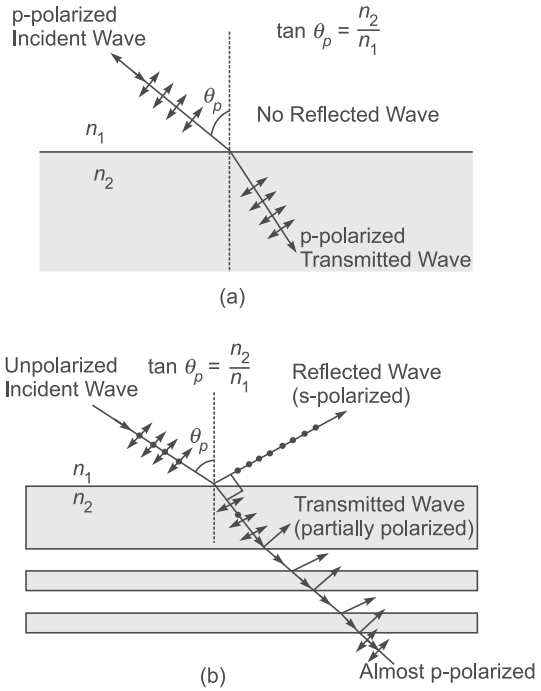


Fig. 22.9 (a) If a p-polarized wave (**E** in the plane of incidence) is incident on the interface of two dielectrics with the angle of incidence equal to q_p (= $\tan^{-1} n_2/n_1$) then the reflection coefficient is zero. (b)If an unpolarized beam is incident at Brewester's angle, the reflected beam is plane polarized whose electric vector is perpendicular to the plane of incidence. The transmitted beam is partially polarized and if this beam is made to undergo several reflections, then the emergent beam is almost plane polarized with its electric vector in the plane of incidence.

22.3.3 Polarization by Double Refraction

In Sections 22.5 and 22.9 we will discuss the phenomenon of double refraction and will show that when an unpolarized beam enters an anisotropic crystal like calcite, it splits up into two linearly polarized beams (see Fig. 22.13). If by some method, we could eliminate one of the beams, then we would obtain a linearly polarized beam.

A simple method for eliminating one of the beams is through selective absorption; this property of selective absorption is known as dichroism. A crystal like tourmaline has different coefficients of absorption for the two linearly

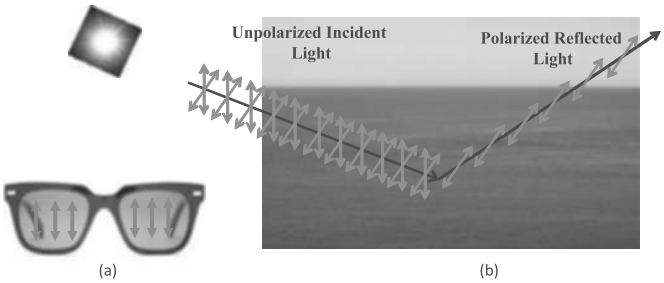


Fig. 22.10 (a) A (commercially available) polarized sunglass blocks the horizontal component and allows only the vertical component to pass through. (b) If the sunlight is incident on the water surface at an angle close to the Brewster angle, then the reflected light will be almost polarized and if we now wear polarized sunglasses, the glare, i.e., the light reflected from the water surface will not be seen. Polarized sunglasses are often used by fishermen to remove the glare on the surface and see the fish inside water. See also Fig. 27 in the prelim pages.



Fig. 22.11 (a) A photograph on the road with ordinary glasses. (b) If we use polarized lenses, the glare can be considerably reduced. Photographs adapted from www.esaver.com.my/index.php?option=com_content&view=article&id=95&Itemid=220. See also Fig. 28 in the prelim pages.

polarized beams into which the incident beam splits up. Consequently, one of the beams gets absorbed quickly and the other component passes through without much attenuation. Thus, if an unpolarized beam is passed through a tourmaline crystal, the emergent beam will be almost linearly polarized (see Fig. 22.14).

Another method for eliminating one of the polarized beams is through total internal reflection. We will show in Sections 22.5 and 22.10 that the refractive indices corresponding to the two beams are different. If one can sandwich a layer of a material whose refractive index lies between the two, then for one of the beams, the incidence will be at a rarer medium and for the other it will be at a denser medium. This principle is used in a Nicol prism which



Fig. 22.12 If the sunlight is incident on the water surface at an angle close to the Brewster angle, then the reflected light will be almost polarized. (a) If the polaroid allows the (almost polarized) reflected beam to pass through, we see the glare from water surface. (b) The glare can be blocked by using a vertical polarizer and one can see the inside of the water. Figure adapted from the website http://polarization.com/water/water.html created by Dr J Alcoz; used with permission of Dr. Alcoz. A color photo appears as Fig. 29 in the prelim pages.

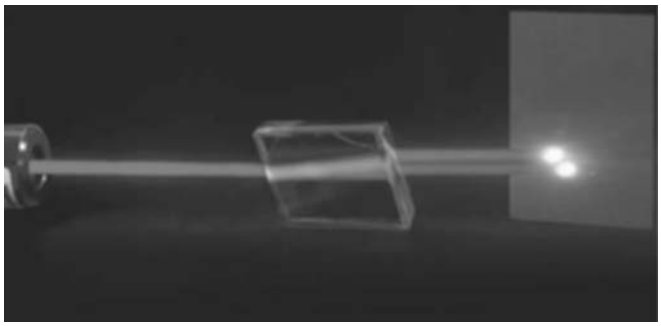


Fig. 22.13 When an unpolarized light beam is incident normally on a calcite crystal, it usually splits up into two linearly polarized beams. Photograph courtesy Professor V Lakshminarayanan and adapted from Ref. 22.16. A color photo appears as Fig. 30 in prelim pages.

consists of a calcite crystal cut in such a way that for the beam, for which the sandwiched material is a rarer medium, the angle of incidence is greater than the critical angle. Thus, this particular beam will be eliminated by total internal reflection. Figure 22.15 shows a properly cut calcite crystal in which a layer of Canada Balsam has been introduced so that the ordinary ray undergoes total internal reflection. The extraordinary component passes through and the beam emerging from the crystal is linearly polarized.

22.3.4 Polarization by Scattering

If an unpolarized beam is allowed to fall on a gas, then the beam scattered at 90° to the incident beam is linearly polarized. This follows from the fact that the waves propagating